

Direct Measurement of Basin Structure in Equilibrium Reasoning Networks

QHRRN-2 measurement theory, self-contained edition for review — definitions, estimators, interpretations, and headline measurements. 2026-08-12. Companions: Design_Ledger.md (the evidence chain), probe_* tools (implementations), github.com/Aakash-Marthandan/QHRRN.

0. The system in one paragraph

We study small ($\sim 0.08\text{M}$ – 0.45M parameter) convolutional-hierarchical networks that solve ARC-style grid puzzles by relaxation: rather than emitting an answer in one pass, the network repeatedly refines a continuous answer state until it stops changing, and is trained so that correct answers are attractors (stable fixed points) of this refinement. A puzzle ('task') is specified by a few input $\rightarrow$ output example pairs ('supports') plus a query input; a small per-task program vector is fitted to the supports at test time and conditions the network. This note defines the dynamical system precisely and then the instruments we use to measure its attractor landscape directly — the measurements behind our empirical results.

1. Preliminaries

Grids and tasks. A grid is an $H \times W$ array of cell values from a vocabulary $\mathcal{V}$ of V symbols (colors, plus distinguished blank/void states); grids are placed on a fixed 32×32 canvas. A task instance supplies an input grid x , a ground-truth output g^* , and support pairs from which a program vector is fitted. The discrete answer space is $\mathcal{G}$, the set of grids.
State space. The network's answer register is continuous: one probability distribution over $\mathcal{V}$ per canvas cell,

$$y \in \mathcal{S} := (\Delta_V)^{32 \times 32}, \quad \Delta_V = \{p \in \mathbb{R}^V : p_i \geq 0, \sum_i p_i = 1\}.$$

A second carried latent z (a feature field) evolves jointly with y ; all instruments read y .

The network map. One forward pass of the network f_θ , conditioned on the input x and the program vector v , produces per-cell logits

$$\ell_\theta(y; x, v) \in \mathbb{R}^{32 \times 32 \times V}.$$

The program vector ('boundary program'). $v = e_t \in \mathbb{R}^{d_{task}}$ ($d_{task} = 32$) is a small embedding fitted to the task's support pairs by gradient descent at test time with all network weights frozen (the 'keyhole' fit). It is the only task-specific state; the network weights θ are shared across all tasks.

Learned damping (definition). The relaxation step size is a single learned scalar

$$\eta = \rho + (1 - \rho) \text{sigmoid}(\hat{\eta}), \quad \hat{\eta} \in \mathbb{R} \text{ trained end-to-end, } \rho \in [0, 1) \text{ an optional floor (default } \rho = 0).$$

η controls how far the state moves toward the network's current prediction at each iteration: $\eta \rightarrow 0$ is quasi-static relaxation, $\eta = 1$ is full replacement. It is trained jointly with θ ; across our pretrained models training selects $\eta \approx 0.058$ — independently matching the hand-tuned damping $\lambda = 0.05$ of the concurrent Equilibrium Reasoners architecture (arXiv:2605.21488), a mutual validation neither group engineered.

2. The relaxation dynamics and its basins

Definition (relaxation map). With σ the per-cell softmax, one refinement step is

$$T_{\theta, x, v}(y) = y + \eta (\sigma(\ell_{\theta}(y; x, v)) - y),$$

and T^k denotes k applications. Fixed points satisfy $\sigma(\ell_{\theta}(y^*)) = y^*$.

All instruments use the network's final iteration map (its training-schedule parameter frozen at its terminal value), so we measure the asymptotic dynamical system, not a training transient.

Readout and embedding. The readout $R: S \rightarrow \mathcal{G}$ maps a state to a discrete grid

(per-cell argmax; output extent selected by a learned size head; crop). The embedding

$\iota: \mathcal{G} \rightarrow S$ places a grid as a one-hot state. For exact-match benchmarks

the decision-relevant object is the basin of the readout equivalence class: $R^{-1}(g^*)$ is

exactly the set of states that decode to the correct answer, so 'the basin of g^* ' below

always means the basin of this quotient class under T .

3. The instruments

Definition 1 (retention — does the codeword exist?). With horizon $K = 8$:

$$\text{Ret}(g^*) = \mathbf{1}[R(T^K(\iota(g^*))) = g^* \quad \forall k \leq K].$$

Handed the truth as a state, the map keeps it: g^* is a K -step-stationary point of the

readout orbit — the attractor exists. The count $N = \sum_{\text{pairs}} \text{Ret}$ over a

task set estimates codebook size restricted to the truth set. (Wrong attractors are counted by separate instruments and deliberately excluded here.)

Definition 2 (corruption ladder — how wide is the basin?). The corruption operator

C_{ε} resamples $\lceil \varepsilon |g^*| \rceil$ uniformly chosen cells of g^*

i.i.d. over colors — one draw from the Hamming shell of normalized radius ε . Then

$$\text{Lad}(\varepsilon) = \mathbf{1}[R(T^K(\iota(C_{\varepsilon}(g^*)))) = g^*], \quad \varepsilon \in \{.05, .1, .2, .4\}.$$

$S(\varepsilon) = \sum \text{Lad}(\varepsilon)$ is the survival function of basin radii over

the codebook; the per-answer decoding radius is $\mathbf{r}(g^*) = \sup\{\varepsilon : \mathbf{Lad}(\varepsilon) > 0\}$, and the ratio $S(0.2)/S(0)$ (our 'radius') is the fraction of

existing attractors correctable at radius 0.2. This is the operational analog of a code's

minimum distance via $d \approx 2r$, with two stated qualifications: relaxation decoding admits no

exact Hamming-bound theorem, and uniform draws measure typical-direction rather than adversarial

worst-case radius. Statistics: each Lad is one Bernoulli draw; aggregation over 144

(task, query) pairs gives binomial errors, and corruption masks are seeded identically across

models; so between-model contrasts are paired (exact McNemar tests).

Definition 3 (reachability — can the solver get there?). With $\emptyset$ the all-void grid

and t^* the inference horizon:

$$\text{Ex} = \mathbf{1}[R(T^{t^*}(\iota(\emptyset))) = g^*].$$

Whether the standard cold start lies in the truth basin. Our central dissociation is

$\text{Ret} \gg \text{Ex}$: attractors exist that the deterministic quench does not reach —

existence and reachability are different resources, and most interventions move exactly one of them.

Definition 4 (thermal capture — 'hops'). Langevin relaxation at temperature T_k :

$$y_{k+1} = T(y_k) + \sqrt{2 T_k \eta} \xi_k, \quad \xi_k \text{ Gaussian, per-cell mean-subtracted (simplex-tangent),}$$

with $T_k = T_0$ during an anneal window, then 0 (quench; the $\sqrt{2T\eta}$ scale is the canonical discretization — mis-scaled noise fails silently, a documented failure class in our ledger). A hop is the event that the deterministic endpoint is wrong while some sampled trajectory lands exactly on g^* : a joint probe of basin measure and barrier structure under the annealed-reachable distribution.

Definition 5 (the flux ledger — what does the code cost to run?). The network is a spatial coarse-graining hierarchy; at each scale cut s , information crossing the cut passes through variational channels: each site emits a Gaussian message with learned mean and variance, and

$$I_s = \sum_{\text{sites}} \text{KL}(q(m | \text{state}) \parallel \mathcal{N}(0, I)), \quad I_{\text{total}} = \sum_s I_s .$$

By the data-processing inequality this is an exact upper bound on the nats per episode transmitted through cut s — computed, not estimated, which is how these measurements avoid the mutual-information-estimation pathologies of the information-bottleneck literature. Training may price this rate:

$$\mathcal{L} = \text{CE}(\text{answer}) + \beta \sum_s I_s + \beta_{nl} \sum_s A_s ,$$

where A_s is the identically-constructed ledger for nonlocal (attention) channels and β, β_{nl} are the prices (Lagrange multipliers on the rate). 'Priced' models below have $\beta > 0$; 'free' models $\beta = 0$ with the ledger still measured.

Definition 6 (family-transfer gates). Tasks are drawn from parameterized generator families (RE-ARC). A seeded 25% of families is held out of ALL pretraining (instances and originals — enforced in code). Gate sets: rt (fresh instances of trained families) and rg (instances of never-trained families). Every instrument above, evaluated on rg, measures transfer to genuinely unseen rules; ground truth enters only to score, never any solve path, so the gates are resamplable without evidentiary decay.

4. Parameter dictionary

Symbol	Role	Interpretation
η	learned damping (Def. in Sec. 1)	relaxation step size; learned ≈ 0.058 ($\approx$ EqR's hand-set 0.05)
ρ	damping floor	optional lower bound on η (annealing dial; default 0)
$K = 8$	stationarity horizon	retention = K -step readout invariance (finite-horizon, stated as such)
ε	corruption radius	normalized Hamming radius of the probe shell
$N = S(0)$	retention count	codebook size, truth-restricted
$S(\varepsilon)/S(0)$	radius ratio	code-distance survival curve (typical-case)
$\varepsilon^* \approx 0.2$	pull radius	calibrated correctable radius exploited by the decoding stack
β, β_{nl}	channel prices	rate constraints; the code-distance dial per Law 2
I_s, A_s	per-cut flux	information profile across scales; geometric decay = operational area law
T_0, k	sampler temperature, ensemble size	thermal exploration; hop rate = capture cross-section
e_t, d_{task}	program vector, width 32	the only task-specific state; fitted at test time, weights frozen
rt / rg	trained / held-out family gates	instruments conditioned on rule-never-trained (transfer)

5. Headline measurements (seeded; full tables in the ledger)

Four regularities, each with 2–3 seeds per claim-bearing cell across models from 0.078M to 0.45M parameters:

Law 1 (throat). Priced I_{total} per episode is task-determined and declines mildly with capacity — 785, 657, 635, 602 nats across a 6× parameter range — while free models carry $\sim 150,000$ – $173,000$ nats with no measured benefit on any instrument: the task sets the information; capacity improves the code.

Law 2 (count vs. radius). Codebook size N and radius $S(.2)/S(0)$ have disjoint controls: depth, corpus breadth, and dynamics dials multiply attractors without widening them; pricing (and width, given training budget) widens without multiplying. The landscape is a code with independently tunable parameters.

Law 3 (budget-limited radius). The apparent radius decline at the largest width is removed by doubling training steps (radius 0.14/0.22 at 20k steps $\rightarrow$ 0.33 at 40k; in-distribution radius 0.80, our record) — basin geometry has a compute-optimal-flavored scaling law.

Law 4 (transfer-specific pricing dividend). Priced models beat free ones on never-trained-family retention at every width tested ($16 > 13$, $11.7 > 10$, $11.7 > 5.5$, $16 > 9$ at $d=16/24/32/48$), while in-distribution differences are within seed noise: compression pays precisely where generalization is hardest.

6. Remarks on rigor

(i) All basins are basins of the readout quotient — the decision-relevant object for exact-match tasks; sub-threshold drift within an argmax cell is invisible by design. (ii) Retention is finite-horizon stationarity, not a fixed-point existence proof; the horizon was stress-checked at $K = 16$ with mild decay. (iii) The ladder measures typical-direction radius; an adversarial-corruption variant lower-bounding worst-case distance is registered future work. (iv) Every instrument is oracle-diagnostic: ground truth scores outcomes but never enters a solve path, which is what permits evaluation on reserved distributions without contaminating them. (v) All numbers cited here resolve to dated, pre-registered entries in the public ledger with committed analysis artifacts.